UNIVERSITI MALAYA

NFMN

SIM 1005 / SIN 1002

Fundamentals of Spreadsheets / Introduction to Worksheet

Lecture 7: Numerical Integration

By

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Objectives

- To learn basic Newton-Cotes formula for integration
- To apply the basic Newton-Cotes formulas for integral approximation.
- To find the approximate Newton-Cotes integral solutions using Excel.

Original notes by NFM Noor

Numerical Integration

- Numerical methods provide approximate solutions for complex integral problems.
- Operates on iterative calculations.
- The basic methods are
  1) Trapezoidal rule
  2) Simpson 1/3 rule
  3) Simpson 3/8 rule*

Original notes by NFM Noor

Trapezoidal Rule

Let $f(x)$ be a function continuously defined in the domain $[a, b]$.

The integration of $f(x)$ using Trapezoidal rule is given by

$$I = \int_a^b f(x)\,dx = \frac{b-a}{2}[f(a) + f(b)].$$

Original notes by NFM Noor

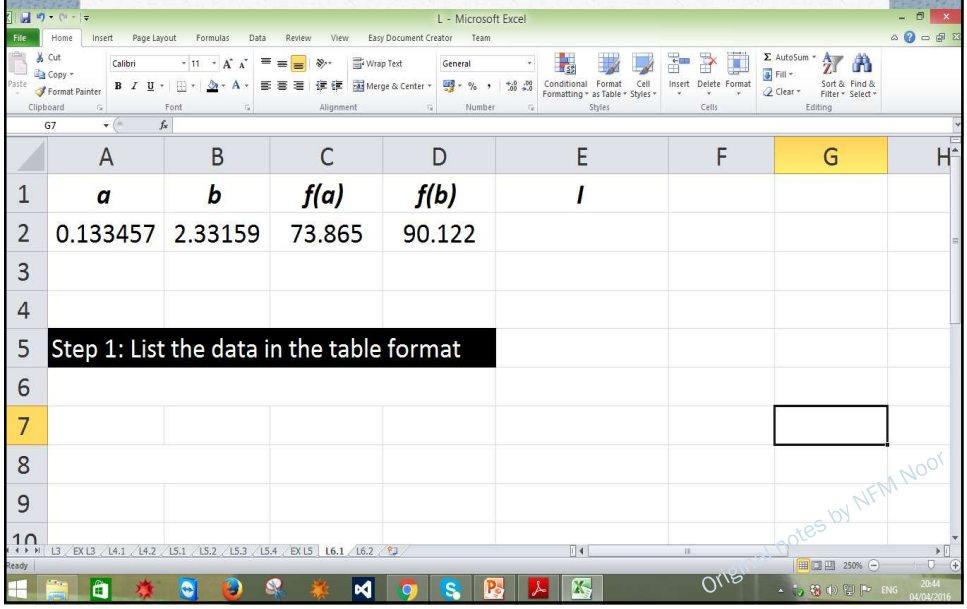
Example 1

Solve $\int_a^b f(x)\,dx$ by using Trapezoidal rule in 6 decimal places if

a	*b*	*f (a)*	*f (b)*
0.133457	2.33159	73.865	90.122

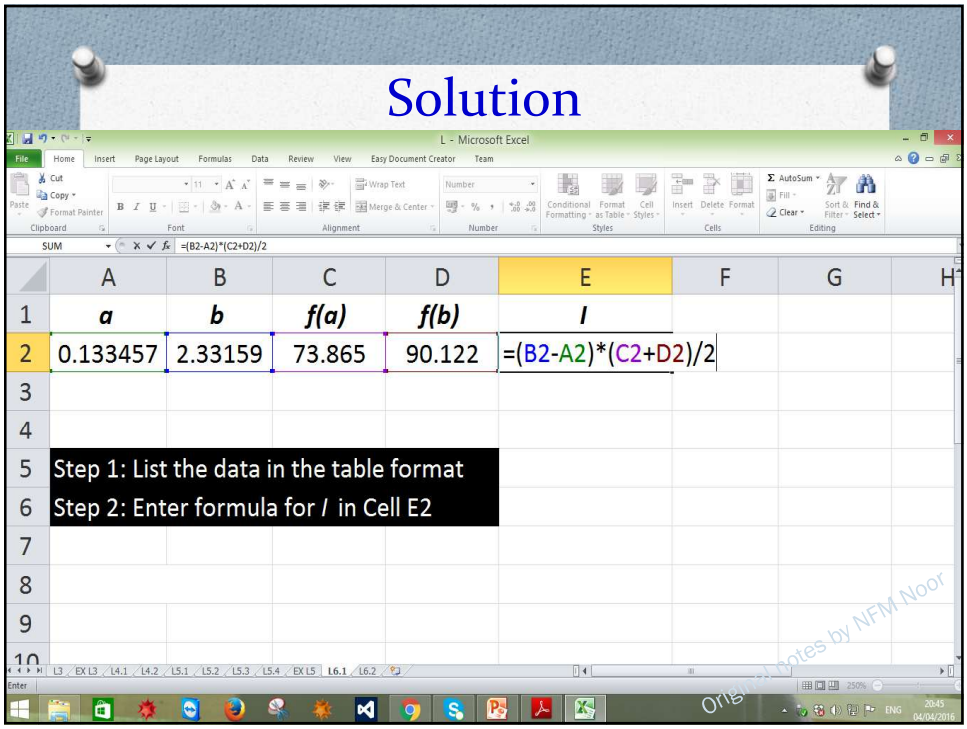
Original notes by NFM Noor

Solution

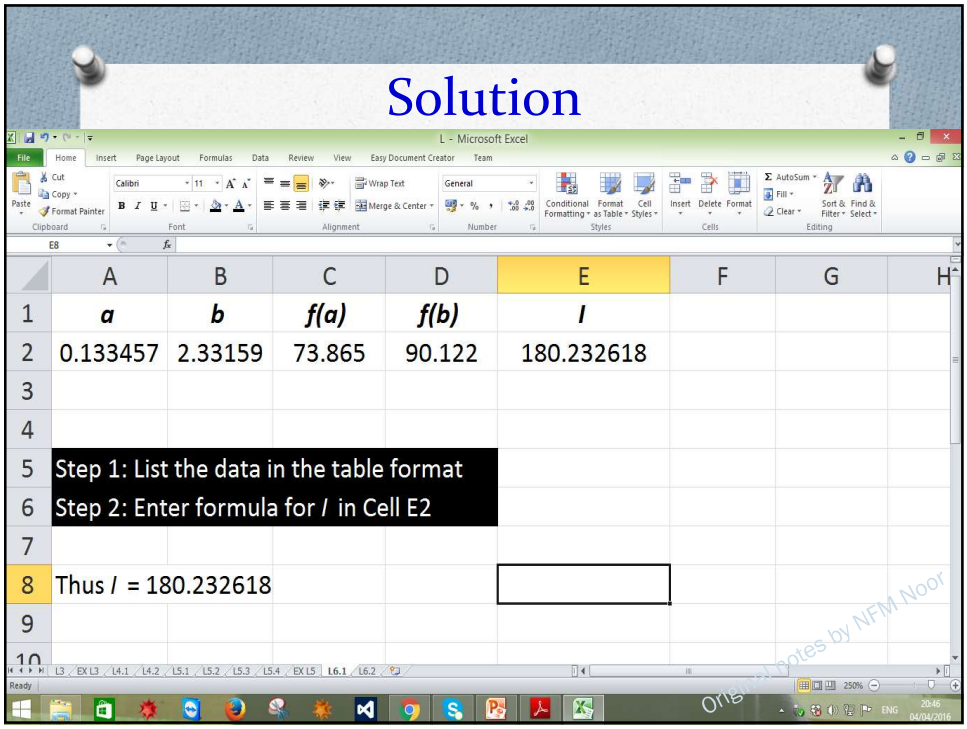


Original notes by NFM Noor

Solution



Solution



Exercise 1

Solve $\int_a^b f(x)\,dx$ by using Trapezoidal rule in 5 decimal places if

a	*b*	*f (a)*	*f (b)*
0	5	56	70
1	4	49	66.78
3.4	4.7	42	44
2	7.6	45.679	81.0324
1.224	2.3	50.008	47.77778
6.7	7.99	70.16	84.0702
6.0	7.5021	65.5	83.12

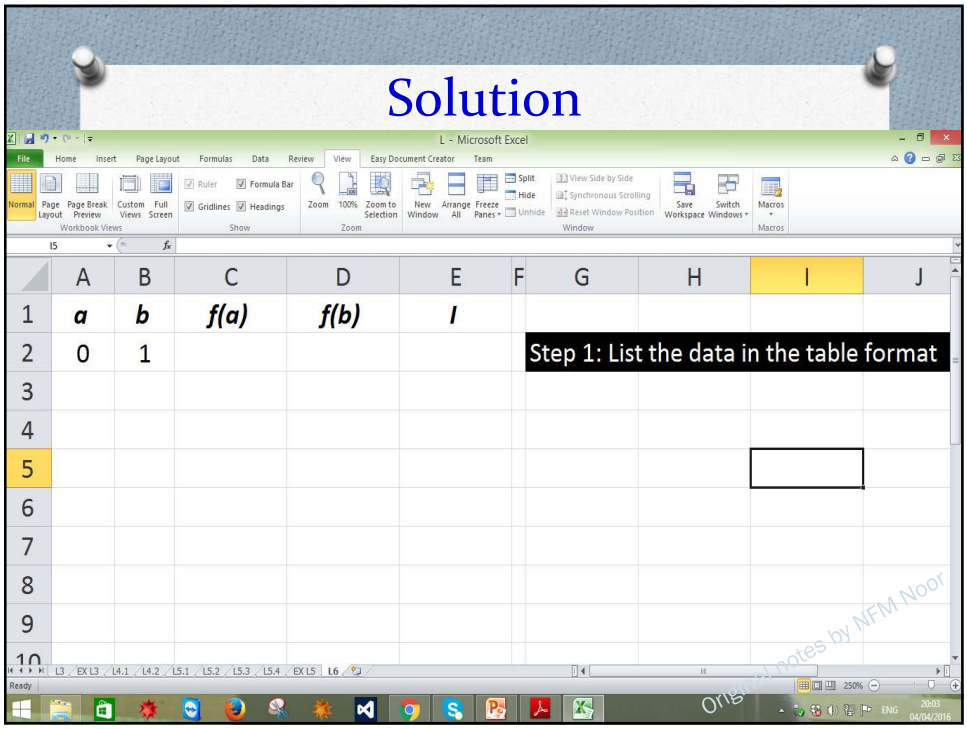
Original notes by NFM Noor

Example 2

Solve $\int_0^1 \frac{5}{2+x}\,dx$ by using Trapezoidal rule in 5 decimal places.

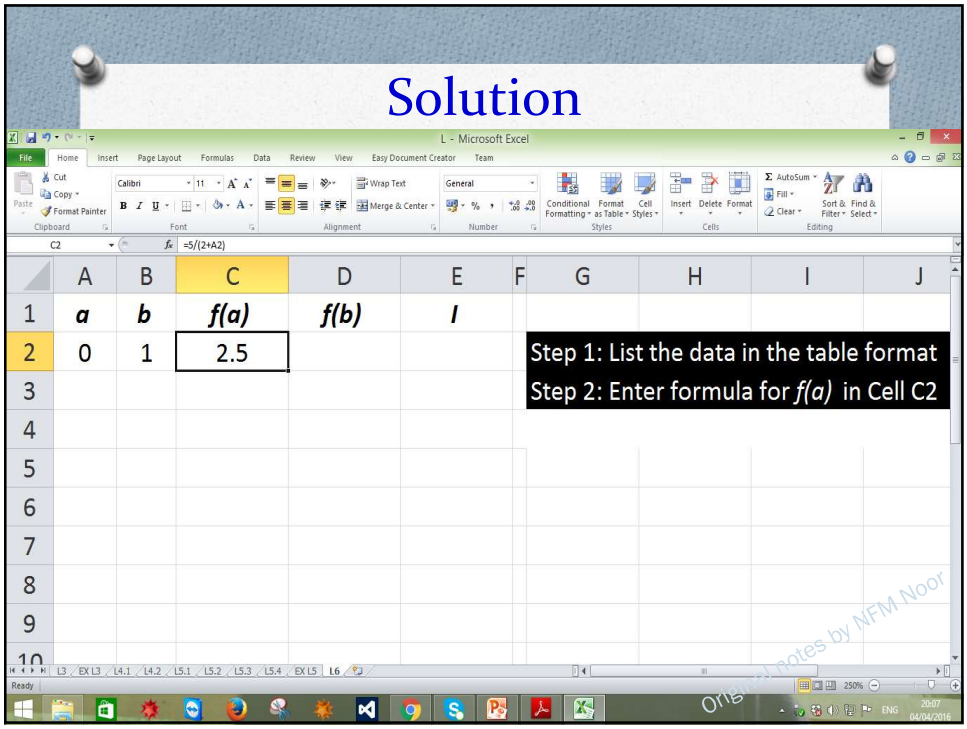
Original notes by NFM Noor

Solution



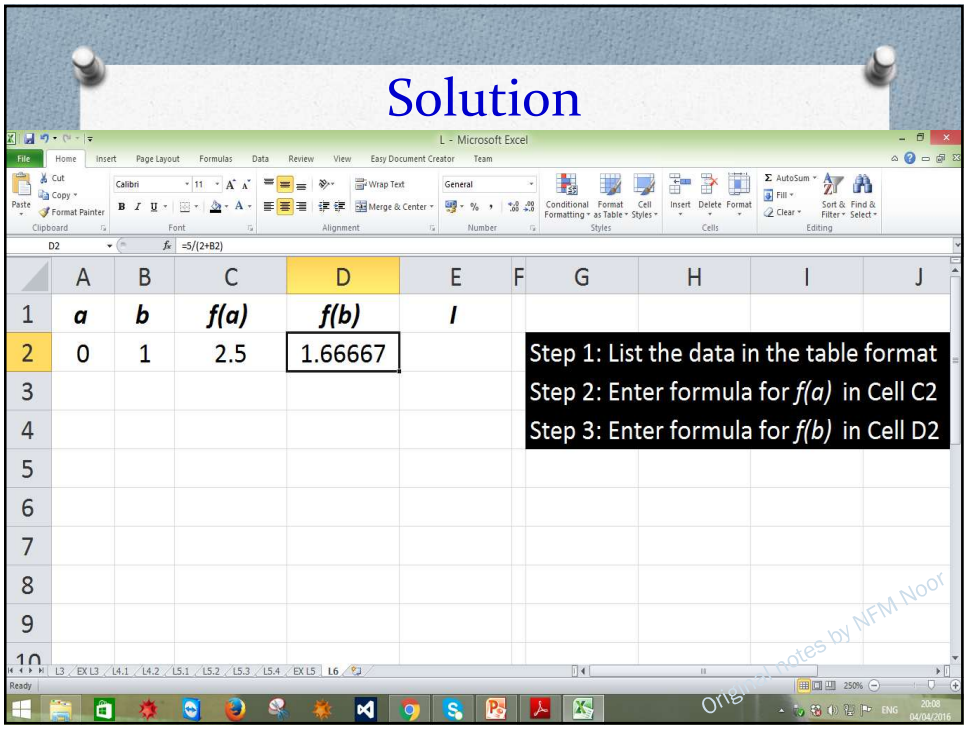
	A	B	C	D	E	F	G	H	I	J
1	*a*	*b*	*f(a)*	*f(b)*	*I*					
2	0	1								
3										
4										
5										
6										
7										
8										
9										
10										

Solution

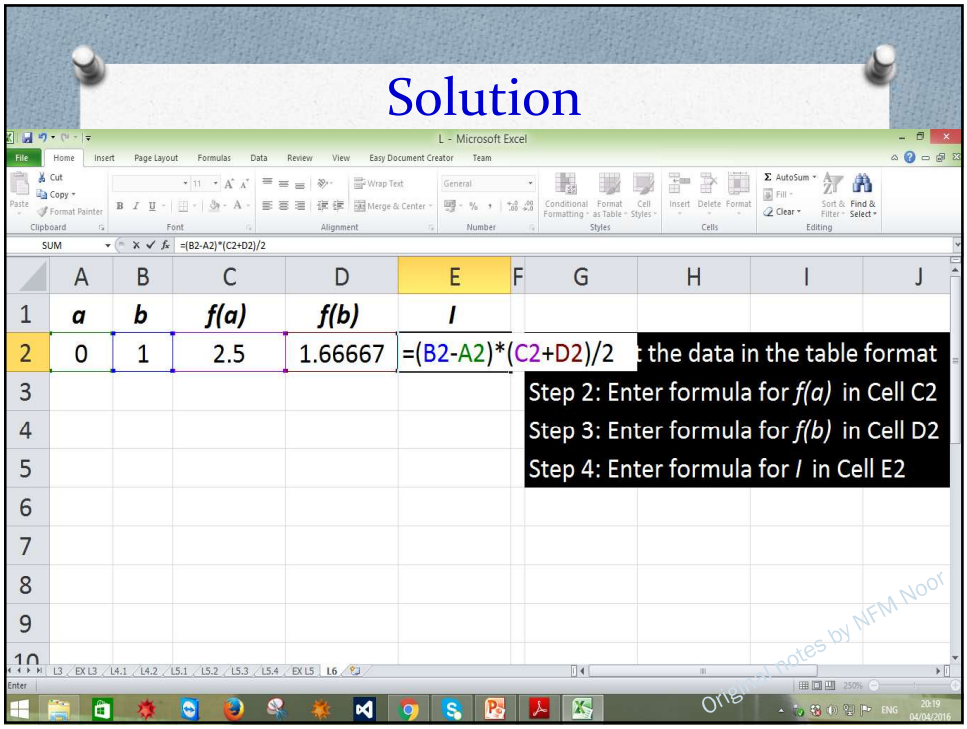


	A	B	C	D	E	F	G	H	I	J
1	*a*	*b*	*f(a)*	*f(b)*	*I*					
2	0	1	2.5							
3										
4										
5										
6										
7										
8										
9										
10										

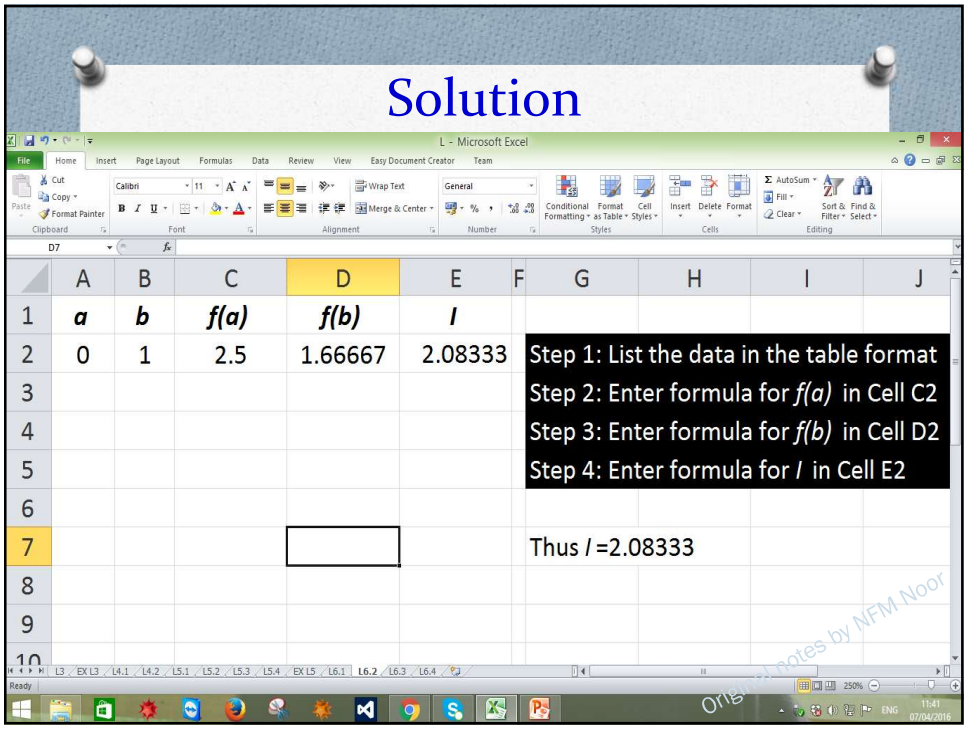
Solution



Solution



Solution



	A	B	C	D	E
1	***a***	***b***	***f(a)***	***f(b)***	***I***
2	0	1	2.5	1.66667	2.08333

Exercise 2

Solve $\int_a^b \frac{5e^x}{7+x^2}\,dx$ by using Trapezoidal rule in 5 decimal places if

a	***b***
0	1
0.05	2
0.25	2.006
0.2771	1.7889
2	5.6
2.4	2.5
0.0014	1.1234

Original notes by NFM Noor

Simpson 1/3 Rule

Let $f(x)$ be a function continuously defined in the domain $[a, b]$.

The integration of $f(x)$ using Simpson 1/3 rule is given by

$$I = \int_a^b f(x)\,dx = \frac{b-a}{6}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right]$$

Original notes by NFM Noor

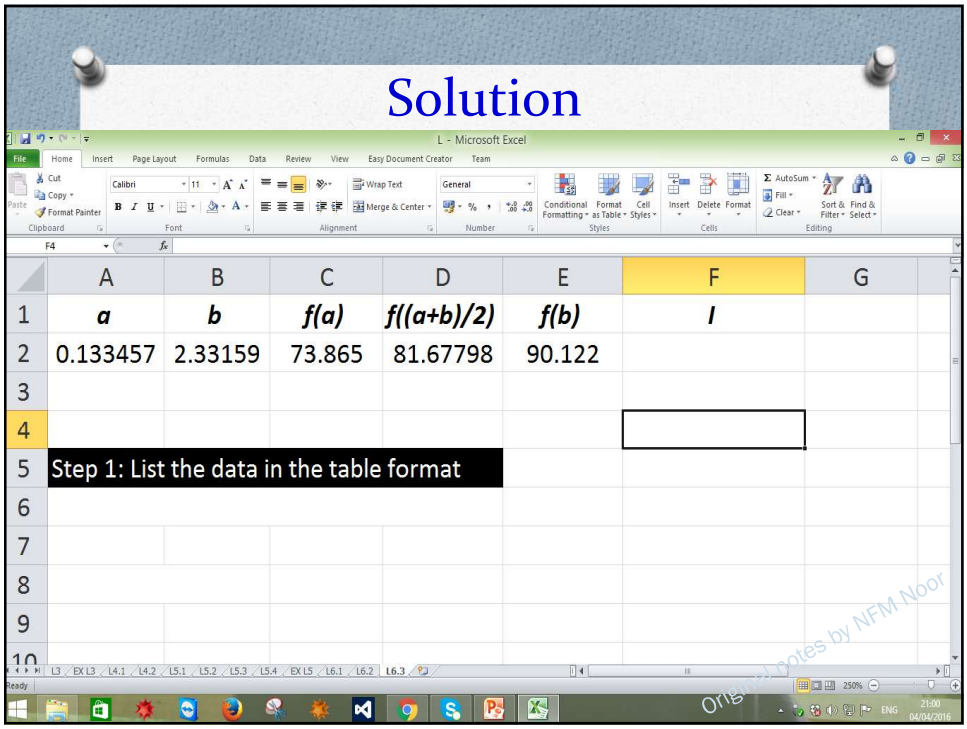
Example 3

Solve $\int_a^b f(x)\,dx$ by using Simpson 1/3 rule in 6 decimal places if

a	b	f (a)	f ((a+b)/2)	f (b)
0.133457	2.33159	73.865	81.67798	90.122

Original notes by NFM Noor

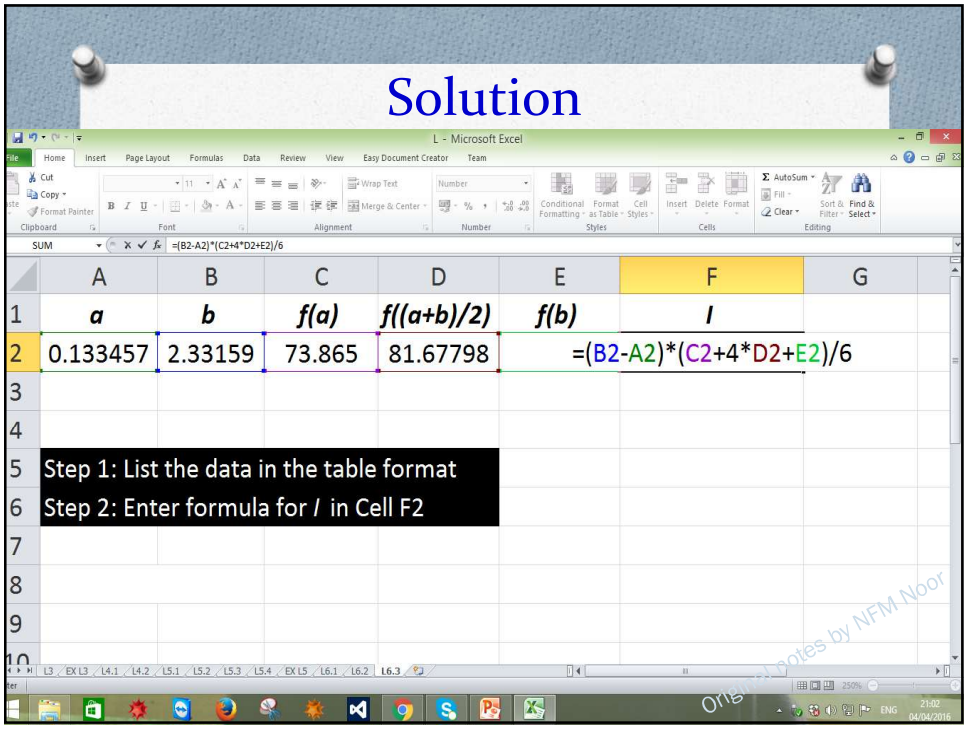
Solution



	A	B	C	D	E	F	G
1	***a***	***b***	***f(a)***	***f((a+b)/2)***	***f(b)***	***I***	
2	0.133457	2.33159	73.865	81.67798	90.122		
3							
4							
5	Step 1: List the data in the table format						
6							
7							
8							
9							
10							

Original notes by NFM Noor

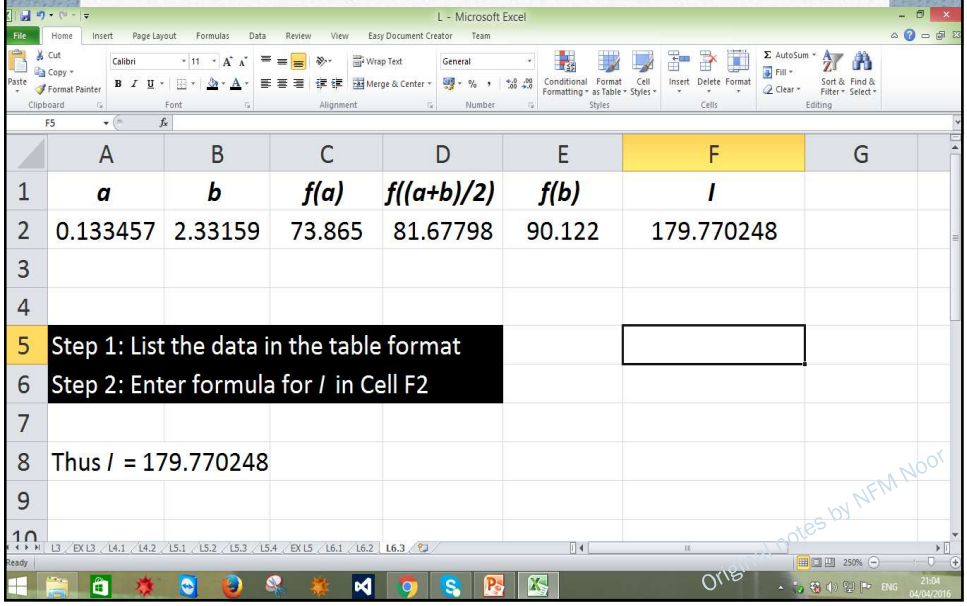
Solution



	A	B	C	D	E	F	G
1	***a***	***b***	***f(a)***	***f((a+b)/2)***	***f(b)***	***I***	
2	0.133457	2.33159	73.865	81.67798		=(B2-A2)*(C2+4*D2+E2)/6	
3							
4							
5	Step 1: List the data in the table format						
6	Step 2: Enter formula for *I* in Cell F2						
7							
8							
9							
10							

Original notes by NFM Noor

Solution



	A	B	C	D	E	F	G
1	*a*	*b*	*f(a)*	*f((a+b)/2)*	*f(b)*	*I*	
2	0.133457	2.33159	73.865	81.67798	90.122	179.770248	
3							
4							
5	Step 1: List the data in the table format						
6	Step 2: Enter formula for *I* in Cell F2						
7							
8	Thus *I* = 179.770248						
9							
10							

Exercise 3

Solve $\int_a^b f(x)\,dx$ by using Simpson 1/3 rule in 5 decimal places if

a	*b*	*f (a)*	*f ((a+b)/2)*	*f (b)*
0	5	56	61.208	70
1	4	49	54.5532	66.78
3.4	4.7	42	43	44
2	7.6	45.679	67.925	81.0324
1.224	2.3	50.008	49.002	47.77778
6.7	7.99	70.16	76.826	84.0702
6.0	7.5021	65.5	77.276	83.12

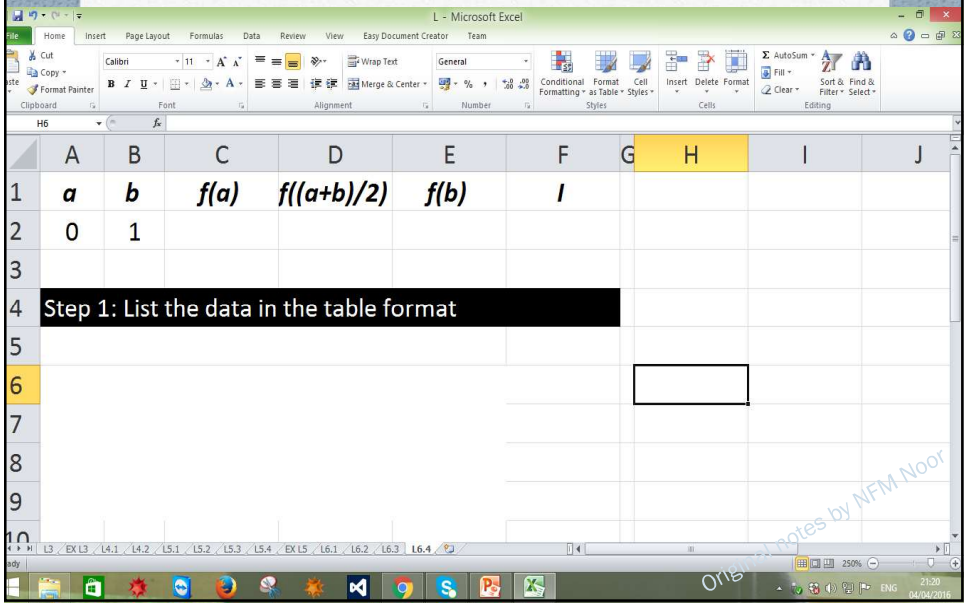
Original notes by NFM Noor

Example 4

Solve $\int_0^1 \frac{5}{2+x}\,dx$ by using Simpson 1/3 rule in 5 decimal places. Calculate the relative approximate error between the solutions of Simpson 1/3 and Trapezoidal rules.

Original notes by NFM Noor

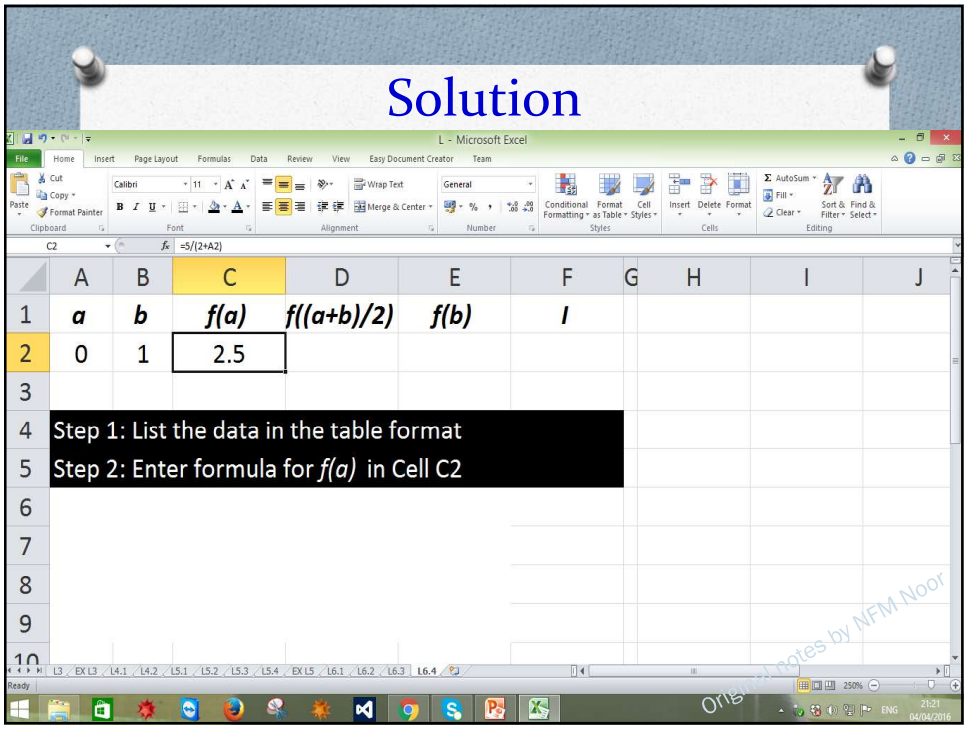
Solution



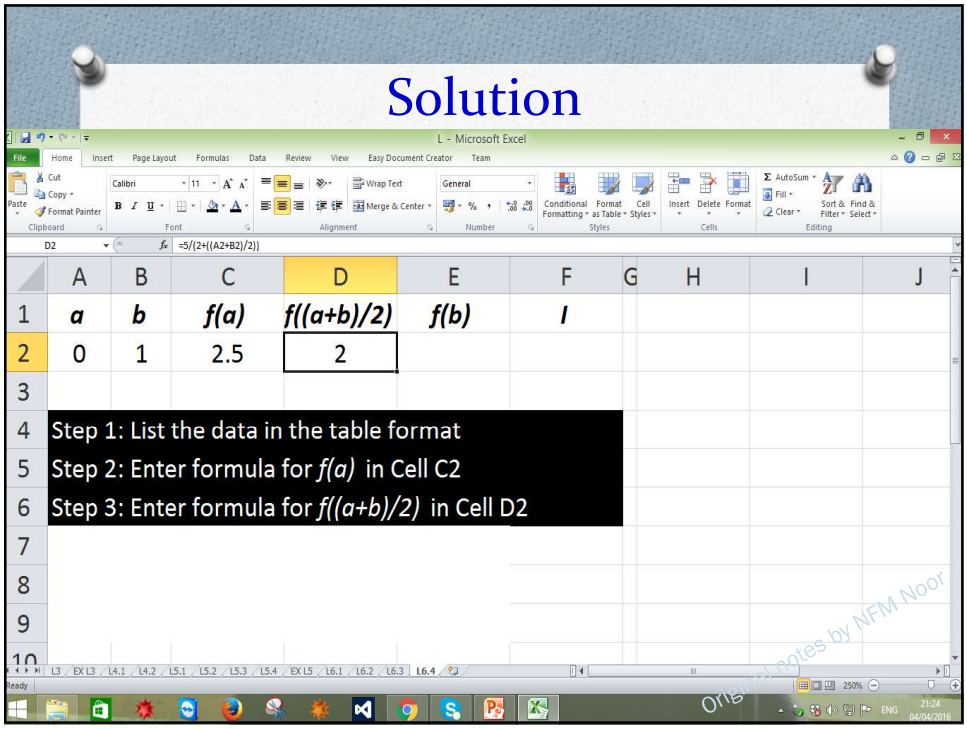
	A	B	C	D	E	F
1	*a*	*b*	*f(a)*	*f((a+b)/2)*	*f(b)*	*I*
2	0	1				

Original notes by NFM Noor

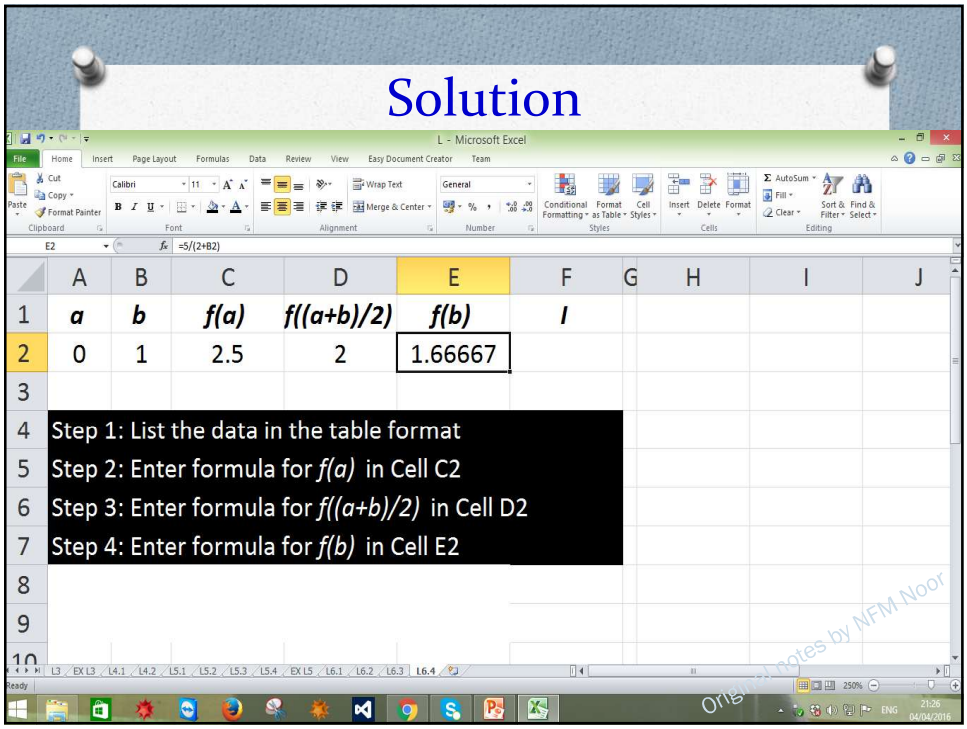
Solution



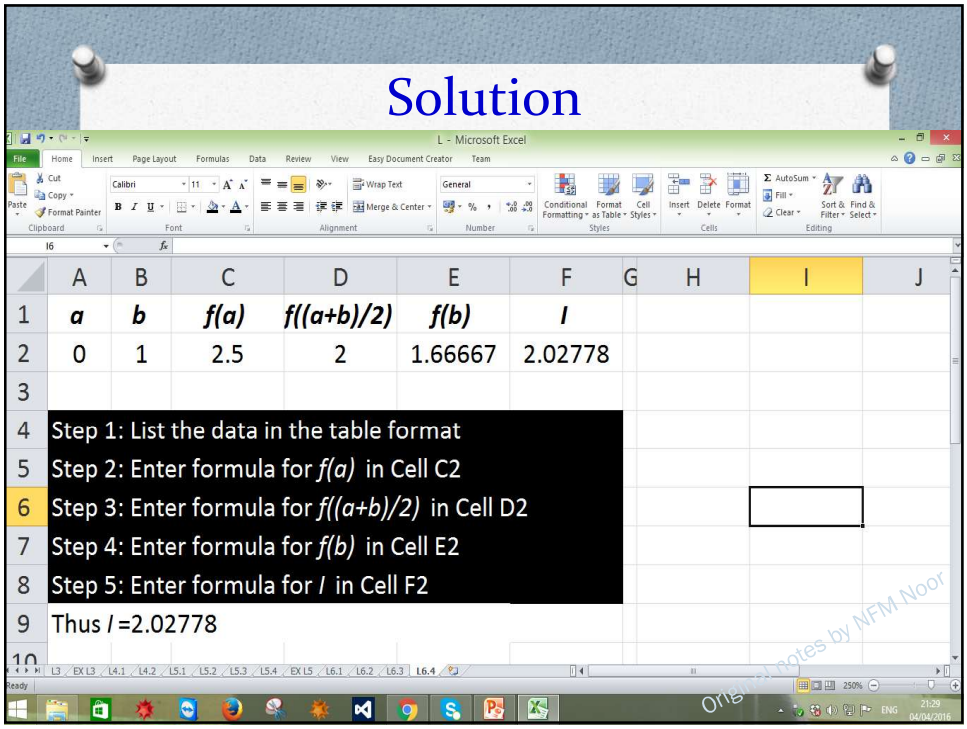
Solution



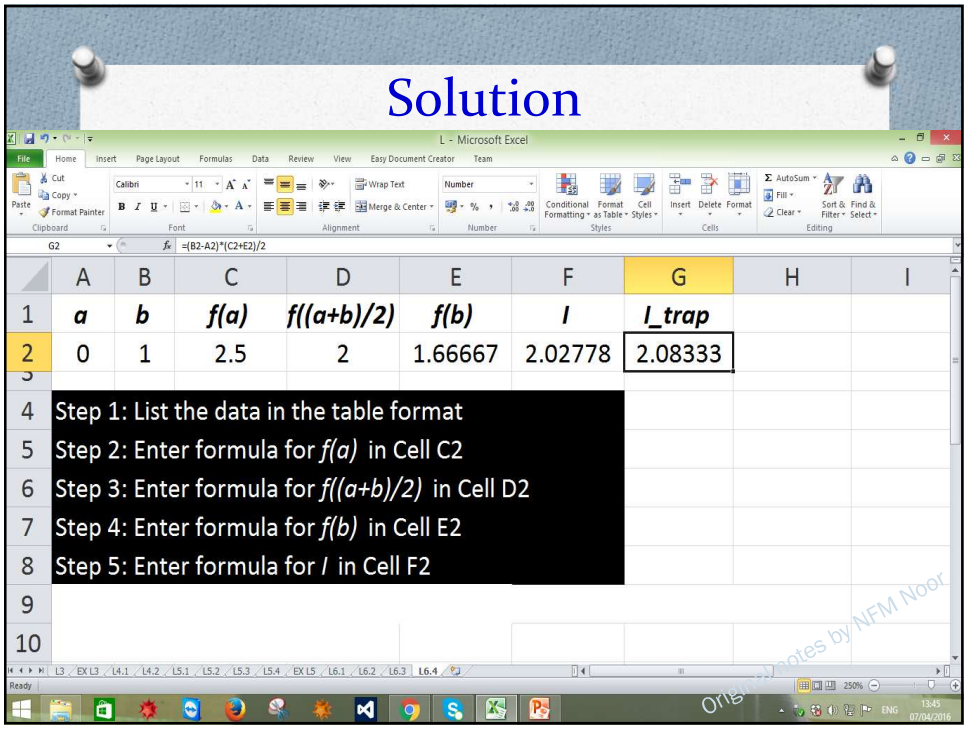
Solution



Solution



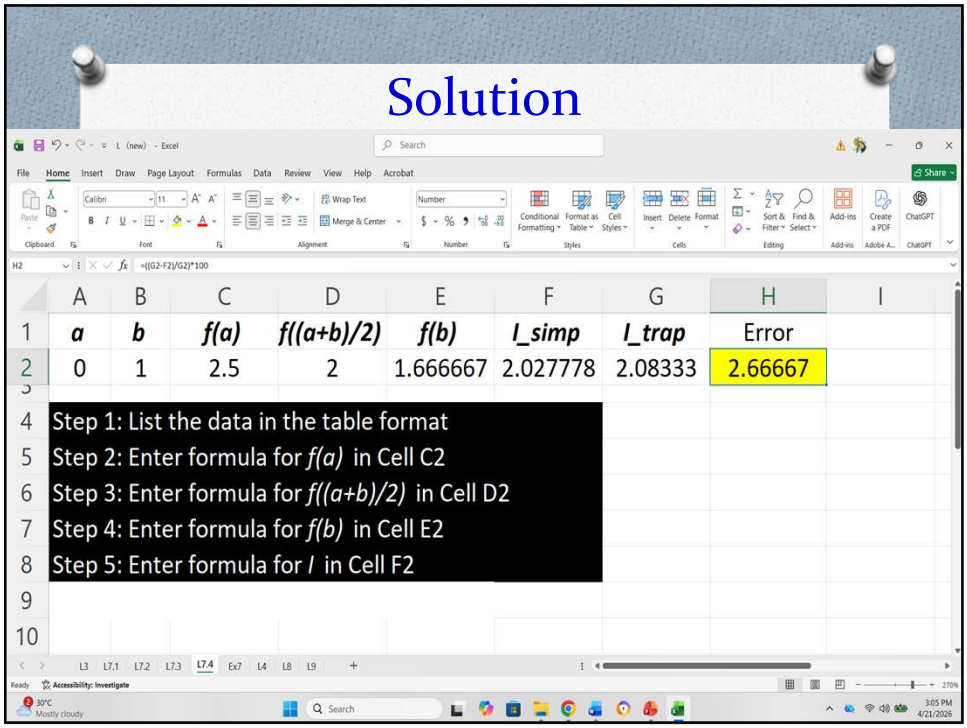
Solution



	A	B	C	D	E	F	G	H	I
1	*a*	*b*	*f(a)*	*f((a+b)/2)*	*f(b)*	*I*	*I_trap*		
2	0	1	2.5	2	1.66667	2.02778	2.08333		
4	Step 1: List the data in the table format								
5	Step 2: Enter formula for *f(a)* in Cell C2								
6	Step 3: Enter formula for *f((a+b)/2)* in Cell D2								
7	Step 4: Enter formula for *f(b)* in Cell E2								
8	Step 5: Enter formula for *I* in Cell F2								
9									
10									

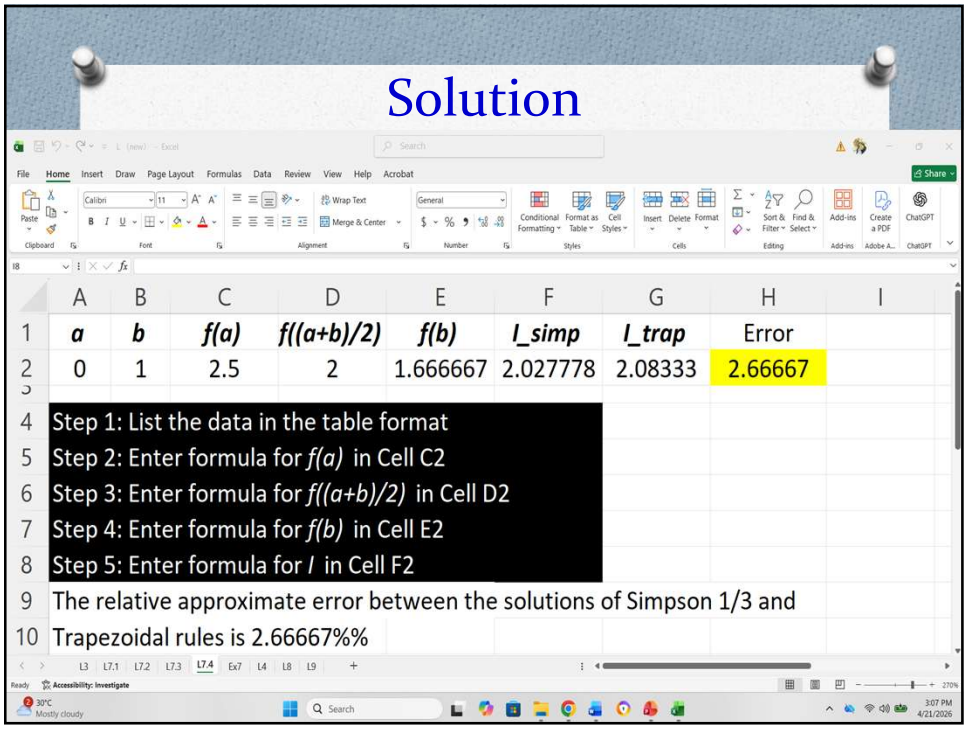
Original notes by NFM Noor

Solution



	A	B	C	D	E	F	G	H	I
1	*a*	*b*	*f(a)*	*f((a+b)/2)*	*f(b)*	*I_simp*	*I_trap*	Error	
2	0	1	2.5	2	1.666667	2.027778	2.08333	2.66667	
4	Step 1: List the data in the table format								
5	Step 2: Enter formula for *f(a)* in Cell C2								
6	Step 3: Enter formula for *f((a+b)/2)* in Cell D2								
7	Step 4: Enter formula for *f(b)* in Cell E2								
8	Step 5: Enter formula for *I* in Cell F2								
9									
10									

Solution



	A	B	C	D	E	F	G	H	I
1	***a***	***b***	***f(a)***	***f((a+b)/2)***	***f(b)***	***I_simp***	***I_trap***	Error	
2	0	1	2.5	2	1.666667	2.027778	2.08333	2.66667	
4	Step 1: List the data in the table format								
5	Step 2: Enter formula for *f(a)* in Cell C2								
6	Step 3: Enter formula for *f((a+b)/2)* in Cell D2								
7	Step 4: Enter formula for *f(b)* in Cell E2								
8	Step 5: Enter formula for *I* in Cell F2								
9	The relative approximate error between the solutions of Simpson 1/3 and								
10	Trapezoidal rules is 2.66667%%								

Exercise 4

Solve $\int_a^b \frac{5e^x}{7+x^2}\,dx$ by using Simpson 1/3 rule in 5 decimal places if

a	***b***
0	1
0.05	2
0.25	2.006
0.2771	1.7889
2	5.6
2.4	2.5
0.0014	1.1234

Hence, find the relative approximate error between the solutions of Simpson 1/3 and Trapezoidal rules.

Original notes by NFM Noor

Summary

- Trapezoidal rule, Simpson 1/3 rule and Simpson 3/8 rule are basic Newton-Cotes formulae for numerical integration.
- Numerical integration is an alternative approach to approximate an integral of a function especially when it involves complex terms.
- The numerical integration formulae can easily be iterated using Excel.

Original notes by NFM Noor